

# When Does LLM-Serving Scheduler Adaptation Matter?

## Action Opportunity and Causal Headroom in Production-Derived Replay

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### 1. Introduction

Modern LLM-serving systems adapt to prompt length, decode demand, KV-cache residency, queueing, deadlines, and changing load. This is necessary in many regimes, but it also creates a practical evaluation problem. A policy may look different in its ranking score while producing the same executable batch, admission, or request order as a strong default. Conversely, a different action may exist without improving the operational metric that motivated adaptation.

We therefore ask a prior question to scheduler design: *when is there actually a different useful action to take?* This question complements systems that design better schedulers, migrate or rebalance requests, optimize batching, disaggregate prefill from decode, predict policy choices, or orchestrate resources. Recent performance-evaluation literature emphasizes the necessity of rigorous ML inference scheduling analysis [1] and adaptive cloud resource management [2]. ServeGen demonstrates why production-shaped workloads matter for serving evaluation [3]; LMetric demonstrates that simple KV-capacity and load-aware routing can materially improve latency on real workloads [4]. We do not replace these systems or claim scheduler superiority over them. We measure a different layer of the argument: the transition from resource pressure to canonical executable choice, and then from choice to local causal value.

The paper follows the decomposition

$$\begin{aligned} \text{resource pressure} &\longrightarrow P(D) \longrightarrow P(B_{\text{LAT}} \mid D) \\ &\longrightarrow \text{headroom magnitude} \longrightarrow \text{predictability and deployment.} \end{aligned} \tag{1}$$

Here  $D$  means that at least one canonical alternative action differs from single best scheduler, while  $B_{\text{LAT}}$  means that at least one such action improves one-step single best scheduler-relative mean latency. The last arrow is deliberately left unclaimed: a causal opportunity is not a learned controller.

The contribution is not the observation that pressure can make policies differ. The contribution is the empirical characterization of when a default has no executable alternative in the studied serving replay, which pressure dimensions create action support, whether that support reproduces on fresh windows, whether disagreement states contain default-relative latency headroom, whether the conclusion depends on the objective, and how far a bounded real serving engine supports the same mechanism.

We answer six empirical questions.

**RQ1 – Abundance.** Under faithful production-derived replay, how often do single best scheduler and the six-policy portfolio produce genuinely different canonical executable actions?

**RQ2 – Pressure support.** Which pressure dimensions create executable disagreement, and which do not?

**RQ3 – Fresh support.** Does the agreement-to-disagreement transition reproduce on untouched non-overlapping production-derived windows?

**RQ4 – Causal headroom.** Conditional on disagreement, how often is single best scheduler locally suboptimal for operational latency, and how large is one-step oracle headroom?

**RQ5 – Objective.** Does the conclusion depend on measuring deadline goodput or latency?

**RQ6 – Real system.** Does a bounded real-vLLM deployment show the corresponding pressure/action mechanism, without claiming real causal headroom?

The central result is conditional rather than universal. Native replay is action-null under the abundant-resource baseline. Concrete KV and active sequence pressures create executable alternatives. Among selected untouched fresh disagreement states, most contain positive local mean-latency headroom, but the mean is modest and the opportunity prevalence can be tiny. The real-system result supports the structural pressure/action mechanism only partially, and we make no claim that a selector has learned to exploit it.

### 1.1. Contributions

This paper makes four contributions:

1. An action-opportunity methodology that distinguishes canonical executable disagreement from policy-score diversity or whole-policy ranking differences.
2. A production-derived regime characterization showing action collapse under native resource abundance and disagreement onset under concrete KV or active-sequence pressure, with the support transition repeated on untouched windows.
3. A fresh default-relative one-step causal analysis that confirms positive mean-latency headroom on untouched disagreement states while retaining the full-state opportunity denominator and treating the oracle as an opportunity bound rather than a scheduler.
4. An objective and systems qualification: terminal arrival-normalized weighted goodput can saturate while latency changes, and bounded vLLM evidence supports the pressure and action-differentiation mechanism without supplying real-system causal latency magnitude or deployed-policy evidence.

## 2. Related Work and Positioning

LLM-serving research spans model execution, prefill/decode scheduling, resource management, routing, caching, and cluster control. Orca and vLLM established iteration-level serving and paged KV management [5, 6]. Sarathi-Serve, DistServe, Splitwise, and FastServe study chunking, phase separation, and preemptive scheduling [7, 8, 9, 10]. Llumnix and SOLA adapt scheduling or state-aware service to dynamic conditions [11, 12]; Libra studies flexible request partitioning [13]; Mooncake and Strata emphasize KV-centric or hierarchical context caching [14, 15]; MorphServe changes model precision and

KV capacity under pressure [16]; and DynamoLLM reconfigures cluster resources for performance and energy [17]. BurstGPT and Azure traces provide important workload evidence [18, 19]; the recent longitudinal study by Nixon et al. further emphasizes workload evolution, caching, and load-balancing over a year of production traffic [20].

ServeGen and OpenTela narrow any generic claim that workload realism or operational trace infrastructure is itself novel [3, 21]. LMetric is especially close: it combines a KV-aware indicator with an instance-load indicator, reports production-style workloads, and derives conditions under which its multiplication can fail [4]. These works answer how to build or evaluate better serving mechanisms. Our question is complementary: in a fixed workload/resource regime, do the default and alternatives expose different canonical actions at all, and does forcing one alternative action have default-relative causal value?

For this reason, systems such as ServeGen, Libra, Llumnix, MorphServe, Strata, LMetric, and OpenTela are related work and design context, not unimplemented baselines for a superiority claim. Reproducing each system would be necessary for a paper claiming that a new scheduler beats them. This paper instead asks whether an executable action opportunity and local default-relative headroom exist before building an adaptive selector. The single best scheduler/VBS-style quantities used below are non-deployable references for opportunity measurement, not schedulers proposed for production.

Table 1 records the scope honestly. A “partial” entry means that a work contains related evidence but does not use the paper’s canonical action or one-step estimand.

Our use of single best scheduler/VBS is rooted in algorithm portfolios and selection [22, 23, 24, 25]. The novelty claim is not the envelope itself, nor counterfactual reasoning in general. It is the narrow serving characterization in Eq. 1.

### 3. Methods

#### 3.1. Evidence hierarchy and workload inventory

We report four evidence tiers. Tier 1 is native production-derived trace replay. Tier 2 is production-derived aggregate or window validation. Tier 3 is a controlled transformation of a real trace, such as one-axis resource caps. Tier 4 is synthetic or mechanistic workload construction. RQ1 uses

Table 1: Closest-work differentiation. We do not claim that other systems omit pressure, production-derived traces, or adaptive mechanisms; the distinction is the action-level and default-relative causal measurement question.

| Work               |   | Prod.-<br>derived<br>workloads | Pressure<br>sweep | Proposes<br>system | Canonical<br>disagree-<br>ment | One-step<br>default in-<br>tervention | Real sys-<br>tem          |
|--------------------|---|--------------------------------|-------------------|--------------------|--------------------------------|---------------------------------------|---------------------------|
| ServeGen           |   | Yes                            | No                | Generator          | No                             | No                                    | Workload<br>study         |
| LMetric            |   | Yes                            | Implicit          | Yes                | No                             | No                                    | Yes                       |
| OpenTela           |   | Operational<br>traces          | No                | Control<br>plane   | No                             | No                                    | Yes                       |
| Llumnix<br>SOLA    | / | Selected<br>regimes            | Partial           | Yes                | No                             | No                                    | System<br>evalua-<br>tion |
| Mooncake<br>Strata | / | Cache/resource<br>regimes      | Partial           | Yes                | No                             | No                                    | Yes / op-<br>erational    |
| This work          |   | Yes                            | Yes               | No                 | Yes, $P(D)$                    | Yes, simu-<br>lator only              | Bounded<br>partial        |

Tier 1; RQ2 uses Tiers 2–3; fresh causal headroom uses a preregistered Tier-3 intervention over untouched windows; the real-vLLM result is a bounded system validation rather than a claim of production deployment.

The faithful corpus contains 20 windows each from Azure 2023 code, Azure 2023 conversation, and BurstGPT. Source order and token-length/arrival structure are retained. SLO and class overlays are deterministic simulator inputs; future actual output lengths are hidden from policies. The fresh confirmatory campaign uses 20 untouched windows per workload family for support mapping, with causal labeling limited to the frozen selected regimes. BurstGPT did not contribute a selected fresh causal disagreement regime, which is retained as an explicit heterogeneity limitation.

The historical six-policy portfolio contains full prefill, small-chunk prefill, estimated-service-first (ESTF), weighted fair sharing (WFS), least-laxity first (LLF), and a KV-constrained policy. six-policy is a fixed portfolio for a default-relative characterization, not an exhaustive set of 2026 schedulers.

### 3.2. Canonical action opportunity

Let  $\mathcal{A}_6(s)$  be the set of canonical executable actions produced by the six portfolio policies at state  $s$ , after action deduplication. A canonical action is the simulator-executable admission, batch, and request ordering decision after infeasible or semantically identical variants have been collapsed. The default scheduler is single best scheduler, the fixed single best policy used as the canonical baseline for the frozen replay protocol; it is not refit per state. Let  $a_{\text{singlebestscheduler}}(s)$  be single best scheduler’s action. We define

$$D(s) = \mathbf{1} [\exists a \in \mathcal{A}_6(s) : a \neq a_{\text{singlebestscheduler}}(s)]. \quad (2)$$

The opportunity prevalence is  $P(D) = \sum_s D(s) / |\mathcal{S}_{\text{all}}|$ , where the denominator is every valid single best scheduler decision state, not only states where policy scores differ. A policy ranking change that produces the same admission, batch, and request order is action collapse, not opportunity. An alternative action is any executable member of  $\mathcal{A}_6(s)$  that is not equal to  $a_{\text{singlebestscheduler}}(s)$ .

This definition matters operationally. Whole-policy comparisons can show a latency or goodput difference accumulated over many decisions, but cannot say whether an alternative existed at a particular default state. Conversely, action diversity alone says nothing about usefulness.

### 3.3. One-step causal headroom

For every disagreement state  $s$  and canonical alternative  $a$ , we clone the state, force exactly one action, and return to single best scheduler for the continuation. Denote the resulting continuation value by  $Q_{\text{singlebestscheduler}}(s, a)$ . This is a local counterfactual, not selector performance: it estimates the one-step effect of changing the current action while holding the subsequent default policy fixed.

For the fresh primary latency endpoint, continuation mean latency includes active, queued, and future requests belonging to the continuation, excluding requests completed before  $s$ . We define

$$A_{\text{LAT}}(s, a) = L_{\text{singlebestscheduler}}(s) - L_{\text{CF}}(s, a), \quad H_{\text{LAT}}(s) = \max \left( 0, \max_a A_{\text{LAT}}(s, a) \right), \quad (3)$$

where positive values reduce mean latency. The beneficial-state indicator is  $B_{\text{LAT}}(s) = 1$  iff  $\max_a A_{\text{LAT}}(s, a) > 0$ . We report  $P(B_{\text{LAT}} | D)$  and mean  $H_{\text{LAT}}$  separately from  $P(D)$ . The conceptual aggregate opportunity is therefore

$$P(D) \times P(B_{\text{LAT}} | D) \times \text{effect magnitude}, \quad (4)$$

not an exact deployable-performance equation. A high conditional benefit can coexist with a tiny overall opportunity.

The confirmatory mean uses a faithful-window clustered bootstrap with 2,000 replicates and seed 20260920. The cluster unit is the original fresh trace window, identified by workload source and window index, contributing disagreement states, so multiple states from one window do not become independent experimental units. Its verdict requires positive mean headroom and a clustered 95% lower endpoint above zero.

A resource-pressure regime is a workload window plus one frozen one-axis intervention, such as a KV-capacity cap, active-sequence cap, or arrival-rate scaling factor. “Support” and “opportunity” refer only to the existence of executable non-single best scheduler actions under that regime; usefulness requires the separate causal latency test above.

### 3.4. *Objective audit*

The historical objective is arrival-normalized weighted goodput,

$$\text{arrival-normalized weighted goodput} = \frac{\sum_i w_i \mathbf{1}[i \text{ completes by its SLO}]}{\sum_i w_i}. \quad (5)$$

The earlier’s all-terminal-arrival-normalized weighted goodput result is retained as an objective audit, not as fresh latency evidence. The latency study uses Eq. 3; its mean-latency result is confirmatory only for the fresh untouched causal universe.

## 4. Results: Native Action Opportunity

Under the frozen abundant-resource baseline, all three production-derived workloads were canonically action-null (Figure 1 summarizes the evidence chain; Figure 2 gives the rates). Azure code had 0 disagreements in 99,992 single best scheduler decision states; Azure conversation had 0/461,985; and BurstGPT had 0/440,461. The corresponding maximum KV utilizations were 0.002716, 0.001728, and 0.003802, with no active-cap binding states. Policy-ranking proxy differences existed in a small number of states, but canonical actions collapsed. This is not evidence that adaptive scheduling never matters; it is evidence that this resource regime offered no different executable six-policy action.

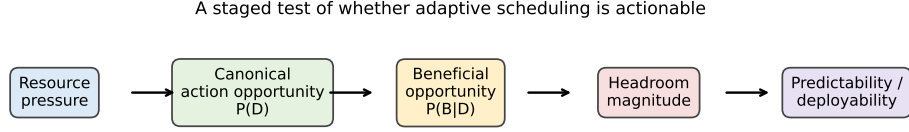


Figure 1: The paper’s evidence chain. A predictor is downstream of both action opportunity and causal value; deployment benefit is not assumed by the characterization.

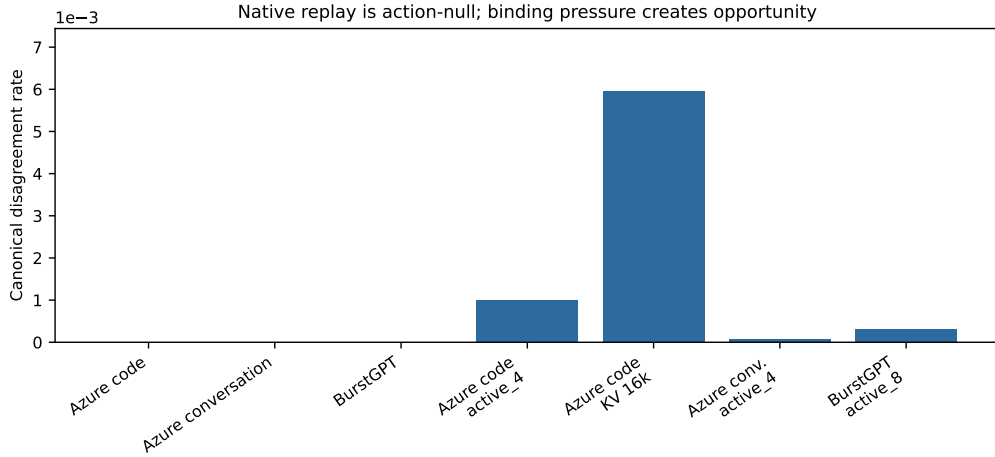


Figure 2: Native production-derived replay is action-null. Selected pressure conditions create nonzero canonical disagreement, with rates depending on workload and pressure axis. This is support for adaptive action selection, not evidence of selector benefit by itself.

## 5. Results: Resource-Regime Transition

The Resource-Regime map evaluated 1,080 one-axis conditions: 20 windows across three workload families, with active-sequence, KV-capacity, and arrival pressure axes (Figure 3; Table 2). There were 926 valid unconstrained, 38 partially constrained, 96 strongly constrained, and 20 horizon-invalid conditions. Across valid conditions, 11,328 canonical disagreement states were observed.

Arrival scaling through  $8\times$  remained action-null for all three workload families. Active-sequence pressure created disagreement at active-cap 8 or 4 depending on workload; KV pressure created disagreement at the first binding or near-binding capacities. For Azure code, the first active-cap dis-

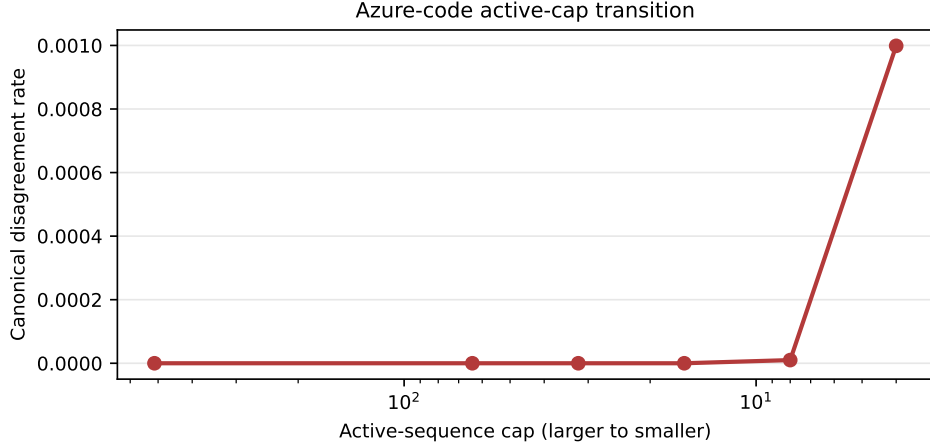


Figure 3: Azure-code active-sequence transition. Agreement persists until the active cap binds the trace, after which canonical disagreement increases.

agreement appeared at active-sequence cap = 8 and became sustained at active-sequence cap = 4. KV disagreement began at KV capacity = 16,000. Azure conversation first disagreed at active-sequence cap = 4 and showed KV disagreement at KV capacity = 16,000; BurstGPT showed active disagreement at active-sequence cap = 16, sustained at active-sequence cap = 8, and KV disagreement at KV capacity = 16,000.

The transition (Table 2) is the main regime result: pressure does not automatically create useful diversity, but concrete resource binding can turn a collapsed action set into a nontrivial one. In the studied traces, arrival-rate scaling alone did not create support through the tested range; tight KV capacity and active-sequence limits were the operative dimensions.

## 6. Results: Fresh Support and Causal Latency Headroom

The fresh support mapping used untouched windows and the frozen pressure protocol. The causal labeling plan selected five Azure code/conversation active/KV regimes and 720 disagreement states. BurstGPT contributed no selected causal regime, an important limitation rather than a missing result.

The confirmatory intervention found 590 beneficial states (Table 3; Figure 4), giving

$$P(B_{\text{LAT}} \mid D) = 590/720 = 0.8194444444.$$

Table 2: Pressure-to-disagreement transition summary. “Sustained” means at least two windows with disagreement.

| Workload           | Axis       | First binding            | First disagreement       |                          |
|--------------------|------------|--------------------------|--------------------------|--------------------------|
| Azure code         | Active cap | active-sequence cap = 8  | active-sequence cap = 8  | active-sequence cap = 8  |
| Azure code         | KV cap     | KV capacity = 16,000     | KV capacity = 16,000     | KV capacity = 16,000     |
| Azure conversation | Active cap | active-sequence cap = 4  | active-sequence cap = 4  | active-sequence cap = 4  |
| Azure conversation | KV cap     | KV capacity = 8,000      | KV capacity = 16,000     | KV capacity = 16,000     |
| BurstGPT           | Active cap | active-sequence cap = 16 | active-sequence cap = 16 | active-sequence cap = 16 |
| BurstGPT           | KV cap     | KV capacity = 16,000     | KV capacity = 16,000     | KV capacity = 16,000     |

Table 3: Fresh causal characterization.  $P(D)$  uses all valid single best scheduler decision states;  $P(B_{\text{LAT}} | D)$  uses only the disagreement states. Per-regime source-window counts report distinct fresh trace windows with at least one disagreement state in that regime. The same source window may contribute to multiple regimes, so the row counts are not additive; the 36 distinct workload-window pairs form the clusters used by the bootstrap.

| Workload/regime                             | States | $P(D)$   | $P(B_{\text{LAT}}   D)$ | Mean $H_{\text{LAT}}$ (ms) |
|---------------------------------------------|--------|----------|-------------------------|----------------------------|
| Azure code, active-sequence cap = 8         | 10     | 0.000106 | 0.400                   | 0.033                      |
| Azure code, active-sequence cap = 4         | 122    | 0.001290 | 0.525                   | 0.077                      |
| Azure code, KV capacity = 16,000            | 431    | 0.004561 | 0.923                   | 3.227                      |
| Azure conversation, active-sequence cap = 4 | 64     | 0.000108 | 0.484                   | 0.217                      |
| Azure conversation, KV capacity = 16,000    | 93     | 0.000157 | 1.000                   | 0.241                      |
| All selected fresh states                   | 720    | —        | 0.819                   | 1.996                      |

Mean oracle latency headroom was 0.001995791 s (1.9958 ms), with a clustered 95% CI of [0.000190884, 0.003546196] s ([0.1909, 3.5462] ms). The frozen verdict was statistically supported. This is statistically supported but operationally modest. It establishes measurable local default-relative latency opportunity in the disagreement-support regime, not that the magnitude exceeds the engineering cost of adaptation. It is also not the claim that 81.9% of all scheduler decisions can be improved:  $P(D)$  is often below one percent in the selected regimes and the conditional statistic is defined only on disagreement states.

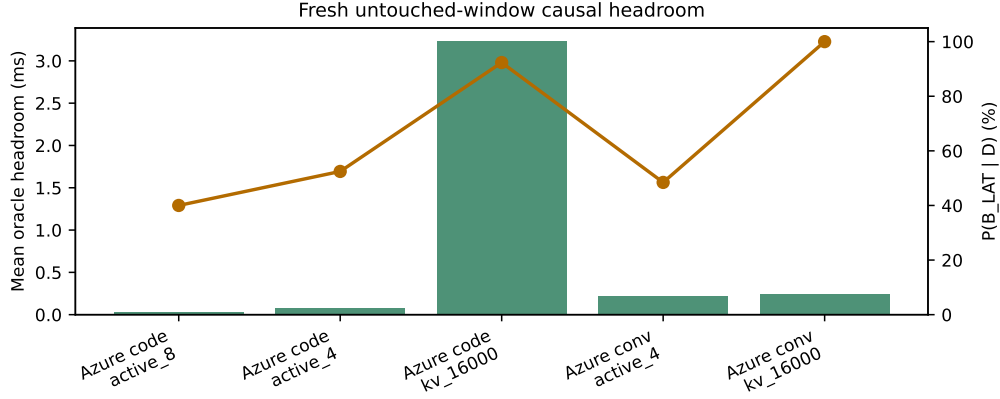


Figure 4: Fresh causal result by selected workload/regime. Bars show mean oracle headroom in milliseconds; the line shows  $P(B_{LAT} | D)$ . The oracle is a non-deployable opportunity bound over observed alternatives.

### 6.1. Opportunity decomposition

Figure 5 makes the distinction between prevalence and value concrete. Azure-code KV capacity = 16,000 has the largest selected  $P(D)$  and the largest mean headroom; active-cap regimes have much smaller  $P(D)$  and correspondingly smaller aggregate opportunity even when some disagreement states are beneficial. A practitioner should therefore measure all three factors in Eq. 4: whether a different action exists, whether it helps conditional on existing, and whether its effect is large enough to matter relative to system noise and control overhead.

The practical reading is deliberately staged. Under abundant resources, this study observed no canonical disagreement and therefore no demonstrated action support for adaptive selection. Under constrained KV or active-sequence regimes, disagreement emerges and creates an executable opportunity. Within those disagreement states, fresh causal evidence shows frequent latency-beneficial alternatives. Opportunity and headroom still do not prove that a deployed adaptive selector is worthwhile.

## 7. Objective Sensitivity: arrival-normalized weighted goodput Saturation

The earlier The earlier pressure study completed with causal integrity and complete branch coverage, but all terminal arrival-normalized weighted good-

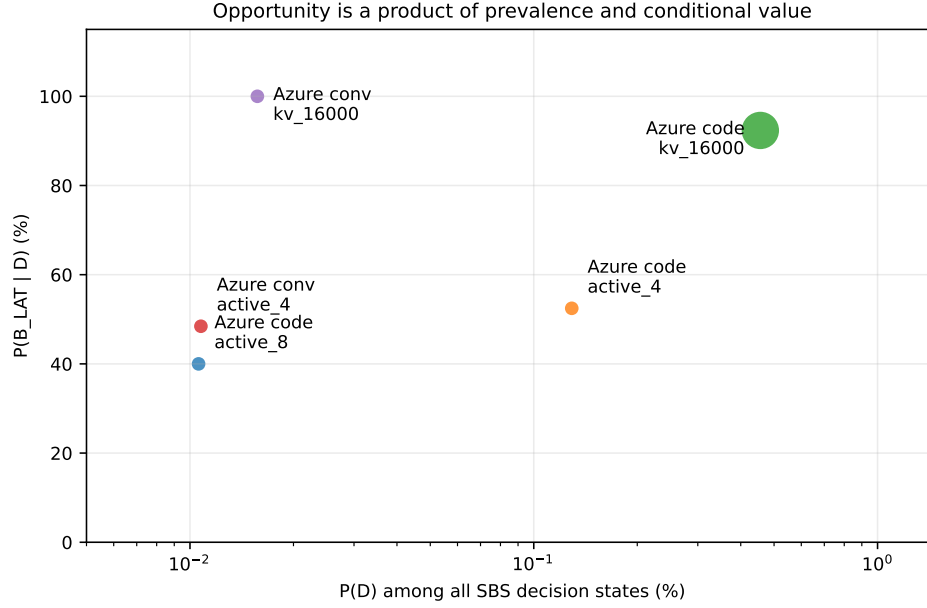


Figure 5: Practitioner regime map. Conditional benefit can be high while opportunity prevalence is small; point size is proportional to mean headroom.

put values were 1.0 and all arrival-normalized weighted goodput advantages were zero. An identifiability audit found no implementation bug: requests completed within deadlines and the objective saturated. Simple sensitivity cases confirmed that arrival-normalized weighted goodput distinguishes missed deadlines or unfinished work, but not different latency or queueing when all requests still meet a loose deadline.

Most counterfactual branches nevertheless changed latency. Thus zero arrival-normalized weighted goodput headroom does not imply zero latency headroom. We do not use the old The earlier post-hoc latency analysis as confirmatory evidence. The fresh untouched study in Section 6 is the confirmatory latency result, with its own pre-frozen primary endpoint and clustered bootstrap.

This is objective sensitivity, not evidence that arrival-normalized weighted goodput is invalid. arrival-normalized weighted goodput is an appropriate deadline-goodput objective for one question; mean latency is a different operational objective. A serving paper should not promote whichever metric yields the more favorable conclusion.

## 8. Bounded Real-vLLM Validation

The bounded real-vLLM validation used one RTX 5060 Ti, vLLM 0.27.1, Qwen2.5-0.5B-Instruct, and deterministic token-shape replay. It completed 40 full/chunked runs and a 20-run native chunked-prefill budget probe. Queueing and scheduler traces were retained. The verdict partially supported means support for the mechanism but not for the full causal or deployment claim. Level 1 is demonstrated only in controlled replay/intervention: at disagreement states, changing the canonical action can expose one-step latency headroom. Level 2 is supported in real vLLM: resource pressure changes observable queue and scheduling behavior. Level 3 is not demonstrated: the paper does not show that a deployed adaptive selector on real vLLM would recover 1.9958 ms or improve end-to-end latency, SLOs, or throughput.

The native budget probe held chunked prefill fixed and varied the token budget. T512 produced 1,680 scheduler steps, 165 mixed prefill/decode steps, and 194 partial-prefill items. T4096 produced 1,536, 55, and 9. Waiting queues reached 7 and 5, respectively, while running sequences reached 4 in both cases. The low-load regime showed a 30.6 ms late-time-to-first-token difference and a 16.3 ms prompt-heavy end-to-end tradeoff; the high-load regime showed a smaller late-time-to-first-token difference and a 23.3 ms prompt-heavy end-to-end tradeoff.

The direct FULL-versus-CHUNKED comparison did not reproduce the simulator’s qualitative reversal (Table 4). KV utilization and preemption metrics were unavailable in the native probe because its metric parser returned empty dictionaries. The real result therefore supports the narrower statement that concrete serving pressure can alter observable scheduling behavior and latency. It is not real-system causal confirmation of Eq. 3.

## 9. Discussion

### 9.1. *When adaptation is justified*

The evidence does not support an anti-adaptive-scheduling conclusion. It supports a staging rule. Under abundant resources, a complex portfolio may collapse to the same action as single best scheduler, making adaptation overhead difficult to justify. Under binding KV or active-sequence pressure, executable alternatives can appear. Some fresh disagreement states contain positive latency headroom, but opportunity prevalence and effect magnitude remain regime-specific.

Table 4: Simulator-to-real-system correspondence. The real-system column supports pressure/action correspondence, not one-step causal headroom.

| Property            | Simulator/replay                                  | Real vLLM                                   |
|---------------------|---------------------------------------------------|---------------------------------------------|
| Low pressure        | Canonical collapse                                | Low queue baseline                          |
| Resource binding    | KV/active caps induce $D$                         | Queue and active-sequence pressure observed |
| Action difference   | single best scheduler/six-policy canonical action | Native token-budget trace signature         |
| Latency consequence | One-step single best scheduler-relative           | Whole-run budget tradeoff                   |
| Exact intervention  | Available                                         | Not available                               |

This matters because scheduler diversity is not scheduler value. A system that produces many distinct actions may still optimize the wrong objective; a system with high conditional benefit may have too few disagreement states to justify a router. The decomposition in Eq. 4 is a practical diagnostic, not a promise of deployment gain. The measured 1.9958 ms mean headroom is therefore evidence that causal opportunity exists in the sampled disagreement-support regime; deciding whether that opportunity exceeds selector latency, engineering complexity, and operational risk is a separate deployment question.

### 9.2. Relation to modern adaptive systems

LMetric, SOLA, Llumnix, MorphServe, Strata, Libra, FastServe, and related systems answer how to adapt serving behavior under dynamic resource and SLO conditions. ServeGen, BurstGPT, Azure traces, OpenTela, and longitudinal trace studies answer how to obtain or operate on realistic workload evidence. Our measurement asks when adaptation is unnecessary because canonical actions collapse, and when a local alternative is causally useful relative to a strong default. The questions are complementary; the present six-policy study does not claim to replace or exhaust modern schedulers, nor does the literature comparison substitute for a benchmark when making a direct performance claim.

### 9.3. Limitations

The supported scope is the studied production-derived trace families, the specific pressure regimes tested, the controlled replay/intervention framework, and the bounded real-vLLM configuration. The six-policy portfolio is not the universe of 2026 schedulers, and different portfolios may expose different support. BurstGPT did not contribute a selected fresh causal regime. Fresh headroom is a mean-latency one-step oracle result, not a learned selector or production improvement; its mean effect is modest and may or may not exceed selector overhead. Fresh non-overlapping windows provide robustness within the trace families, not universal external validity. The real-vLLM validation uses one model, one RTX 5060 Ti GPU, vLLM 0.27.1, and one mechanism family, uses token-shape replay rather than exact private prompts, and supports the pressure/action mechanism only. It has no exact one-step real-system intervention, and the simulator-to-vLLM action mapping is imperfect. arrival-normalized weighted goodput saturation limits conclusions under that objective. Finally, we make no claim over all LLM-serving workloads, GPU architectures, model sizes, scheduler portfolios, serving systems, or forms of resource pressure; no claim of learned predictability, universal adaptive benefit, universal scheduler superiority, or impossibility; and no claim that measured causal headroom automatically justifies deployment.

## 10. Reproducibility and Artifact Scope

The public artifact (the public repository, see Data and Code Availability) records trace provenance, frozen manifests, support grids, fresh non-overlap proofs, causal state and action manifests, source hashes, bootstrap seed 20260920, and the real-vLLM environment. The principal result packages are Phase A native replay, Phase B pressure mapping, fresh confirmatory causal latency, and real-vLLM pressure/action validation. Historical selector and joint-240 artifacts remain available but are not treated as evidence of learned deployment readiness.

## 11. Conclusion

Adaptive LLM serving should be evaluated in stages. Faithful production-derived replay can be action-null under abundant resources; resource pressure can create canonical executable alternatives; and fresh one-step replay can show that some alternatives contain positive local mean-latency headroom.

In this study, the fresh confirmation found 590 beneficial states among 720 disagreement states. Mean oracle headroom was 1.9958 ms, with a positive corrected clustered confidence interval. The result is conditional and modest: the conditional benefit probability does not equate to overall deployment value.

Terminal arrival-normalized weighted goodput saturation demonstrates why objective choice matters, while bounded vLLM evidence provides real-system support for the pressure mechanism and observable action differentiation without claiming real causal headroom or deployment value. This performance-evaluation methodology shows that adaptation should be justified by measured opportunity and causal value; closed-loop and deployed value evaluation remains future work.

### **Declaration of generative AI and AI-assisted technologies in the manuscript preparation process**

During the preparation of this work, the author used ChatGPT by OpenAI, OpenAI Codex, and Google Gemini in order to assist with organization, language refinement, literature-search planning, software development, and workflow support. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

### **Declaration of Competing Interest**

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data and Code Availability

Code, derived research artifacts, experiment configurations, and reproducibility artifacts for this study are available at <https://github.com/SoroushVahidi/llm-serving-heuristic-evolution>. Raw third-party workload traces are obtained from their original upstream sources. A persistent data DOI is currently missing and needs to be archived (DATA\_Doi\_STATUS = MISSING\_NEEDS\_ARCHIVE).

## References

- [1] T. Choudhury, G. Joshi, W. Wang, Job assignment in machine learning inference systems with accuracy constraints, *Performance Evaluation* 167 (2025) 102463. doi:10.1016/j.peva.2024.102463.
- [2] A. Ali, X. Ma, S. Zawad, P. Aditya, I. E. Akkus, R. Chen, L. Yang, F. Yan, Enabling scalable and adaptive machine learning training via serverless computing on public cloud, *Performance Evaluation* 167 (2025) 102451. doi:10.1016/j.peva.2024.102451.
- [3] Y. Xiang, X. Li, K. Qian, W. Yu, E. Zhai, X. Jin, ServeGen: Workload characterization and generation of large language model serving in production, in: *23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI 26)*, USENIX Association, 2026.
- [4] D. Zhang, J. Han, K. Zhang, X. Wei, S. Shen, C. Fang, W. Yu, J. Zhou, R. Chen, LMetric: Simple is better – multiplication may be all you need for LLM request scheduling, in: *20th USENIX Symposium on Operating Systems Design and Implementation (OSDI 26)*, USENIX Association, 2026.

- [5] G.-I. Yu, J. S. Jeong, G.-W. Kim, S. Kim, B.-G. Chun, Orca: A distributed serving system for transformer-based generative models, in: 16th USENIX Symposium on Operating Systems Design and Implementation (OSDI 22), USENIX Association, 2022, pp. 521–538.
- [6] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, I. Stoica, Efficient memory management for large language model serving with PagedAttention, in: Proceedings of the 29th Symposium on Operating Systems Principles (SOSP 2023), ACM, 2023, pp. 611–626.
- [7] A. Agrawal, N. Kedia, A. Panwar, J. Mohan, N. Kwatra, B. S. Gulavani, A. Tumanov, R. Ramjee, Taming throughput-latency tradeoff in LLM inference with Sarathi-Serve, in: 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24), USENIX Association, 2024, pp. 117–134.
- [8] Y. Zhong, S. Liu, J. Chen, J. Hu, Y. Zhu, X. Liu, X. Jin, H. Zhang, DistServe: Disaggregating prefill and decoding for goodput-optimized large language model serving, in: 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24), USENIX Association, 2024, pp. 193–210.
- [9] P. Patel, E. Choukse, C. Zhang, A. Shah, I. Goiri, S. Maleki, R. Bianchini, Splitwise: Efficient generative LLM inference using phase splitting, in: Proceedings of the 51st Annual International Symposium on Computer Architecture, IEEE Computer Society, 2024. doi:10.1109/ISCA59077.2024.00019.
- [10] B. Wu, Y. Zhong, Z. Zhang, S. Liu, F. Liu, Y. Sun, G. Huang, X. Liu, X. Jin, FastServe: Iteration-Level preemptive scheduling for large language model inference, in: 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI 26), USENIX Association, 2026, pp. 57–74.
- [11] B. Sun, Z. Huang, H. Zhao, W. Xiao, X. Zhang, Y. Li, W. Lin, Llumnix: Dynamic scheduling for large language model serving, in: 18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24), USENIX Association, 2024, pp. 173–191.

- [12] K. Hong, X. Li, L. Chen, Q. Mao, G. Dai, X. Ning, S. Yan, Y. Liang, Y. Wang, SOLA: Optimizing SLO attainment for large language model serving with state-aware scheduling, in: *Proceedings of Machine Learning and Systems*, Vol. 7, 2025.
- [13] C. Ruan, Y. Chen, D. Tian, Y. Shi, Y. Wu, J. Li, C. Li, Libra: Flexible request partitioning and scheduling for serving unbalanced and dynamic LLM workloads, in: *23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI 26)*, USENIX Association, 2026.
- [14] R. Qin, Z. Li, W. He, J. Cui, F. Ren, M. Zhang, Y. Wu, W. Zheng, X. Xu, Mooncake: Trading more storage for less computation – a KVCache-centric architecture for serving LLM chatbot, in: *23rd USENIX Conference on File and Storage Technologies (FAST 25)*, USENIX Association, 2025, pp. 155–170.
- [15] Z. Xie, Z. Xu, M. Zhao, Y. An, V. S. Mailthody, S. Mahlke, M. Garland, C. Kozyrakis, Strata: Hierarchical context caching for long context language model serving, *arXiv preprint arXiv:2508.18572* (2025).
- [16] Z. Su, T. Lan, Z. Wang, J. Yang, Y. Cheng, Efficient and workload-aware LLM serving via runtime layer swapping and KV cache resizing, *arXiv preprint arXiv:2506.02006* (2025).
- [17] J. Stojkovic, C. Zhang, I. Goiri, J. Torrellas, E. Choukse, DynamoLLM: Designing LLM inference clusters for performance and energy efficiency, in: *Proceedings of the 2025 IEEE International Symposium on High Performance Computer Architecture (HPCA 25)*, IEEE, 2025, pp. 1348–1362. doi:10.1109/HPCA61900.2025.00102.
- [18] Y. Wang, Y. Chen, Z. Li, X. Kang, Y. Fang, Y. Zhou, Y. Zheng, Z. Tang, X. He, R. Guo, X. Wang, Q. Wang, A. C. Zhou, X. Chu, BurstGPT: A real-world workload dataset to optimize LLM serving systems, in: *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, ACM, 2025, pp. 5831–5841. doi:10.1145/3711896.3737413.
- [19] Microsoft Azure, Azure LLM inference trace 2023, <https://github.com/Azure/AzurePublicDataset/blob/master/>

AzureLLMInferenceDataset2023.md, public LLM inference trace dataset released by Azure (2023).

- [20] W. Nixon, J. Durbin, F. Standhartinger, H. S. Gunawi, J. Yang, A year in LLM serving: Workload evolution, caching and load-balancing, arXiv preprint arXiv:2608.13573 (2026).
- [21] X. Yao, Y. Jiang, I. Badanin, Q. Hu, R. M. Smith, B. Yuan, I. Schlag, E. Yoneki, A. Klimovic, OpenTela: Unifying decentralized computing resources for heterogeneous LLM serving, in: 20th USENIX Symposium on Operating Systems Design and Implementation (OSDI 26), USENIX Association, 2026, pp. 1821–1838.
- [22] J. R. Rice, The algorithm selection problem, *Advances in Computers* 15 (1976) 65–118.
- [23] C. P. Gomes, B. Selman, Algorithm portfolios, *Artificial Intelligence* 126 (1–2) (2001) 43–62.
- [24] L. Xu, F. Hutter, H. H. Hoos, K. Leyton-Brown, SATzilla: Portfolio-based algorithm selection for SAT, *Journal of Artificial Intelligence Research* 32 (2008) 565–606.
- [25] M. Lindauer, H. H. Hoos, F. Hutter, T. Schaub, AutoFolio: An automatically configured algorithm selector, *Journal of Artificial Intelligence Research* 53 (2015) 745–778.